

Khulna University of Engineering & Technology
Department of Building Engineering and Construction Management
 B. Sc. Engineering 2nd Year 1st Term Regular Examination, 2025
Math 2123
 (Mathematics-III)

Full Marks: 210

Time: 3 hrs

- N.B.
- i) Answer any three questions from each section in separate script.
 - ii) Figures in the right margin indicate full marks.
 - iii) Necessary table/ graphs/ charts (if any)

Section – A

1. (a) Define the following matrices with examples: (12)
 (i) Lower triangular matrix, (ii) Symmetric matrix, (iii) Nilpotent matrix,
 (iv) Idempotent matrix.

- (b) Test whether the following matrix A is idempotent or not. (04)

$$A = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}$$

- (c) Express the matrix A as the sum of a symmetric and skew-symmetric matrices. (08)
 Where,

$$A = \begin{bmatrix} 3 & 1 & 0 \\ 5 & 0 & 3 \\ 2 & 4 & 7 \end{bmatrix}$$

- (d) Solve the following system of linear equations by the matrix method: (11)

$$4x - 5y + z = 2$$

$$3x + y - 2z = 9$$

$$x + 4y + z = 5$$

2. (a) Does inverse exist for all matrices? Test whether the following matrix is invertible or not. If possible, find the inverse of the following matrix using elementary (13)

transformation: $\begin{bmatrix} 1 & 20 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{bmatrix}$

- (b) Determine the values of a and b for which the system $3x-2y+z=b$, $5x-8y+9z=3$, $2x+y+az=-1$ (i) has a unique solution, (ii) has no solution, and (iii) has infinite solutions. (12)

- (c) Define linearly dependent and independent vectors. Examine the following vectors for linear dependence: $X_1 = (1, 2, 4)$, $X_2 = (2, -1, 3)$, $X_3 = (0, 1, 2)$. (10)

3. (a) Define rank of a matrix. Reduce the following matrix A to Echelon form, then to canonical form and then to normal form and hence find its rank: where, (20)

$$A = \begin{bmatrix} 2 & -6 & -2 & -3 \\ 5 & -13 & -4 & -7 \\ -1 & 4 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

- (b) Define eigen values and eigen vector. Find the eigen values and eigen vector corresponding one of the eigen values for the matrix (15)

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

4. (a) A particle moves along a curve whose parametric equations are $x = e^{-t}$, $y = 2 \cos 3t$, $z = 2 \sin 3t$, where t is the time. (12)

- (i) Determine the magnitude of the velocity and acceleration at $t = 0$.

- (b) Find the work done in moving an object along a vector $\vec{r} = 3\hat{i} + \hat{j} - 5\hat{k}$ if the applied force is $\vec{F} = 2\hat{i} - \hat{j} - \hat{k}$. (10)
- (c) State Cayley-Hamilton theorem. Verify it for the matrix $A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$ and hence find A^{-1} . (13)

Section - B

5. (a) Write down the sufficient conditions for existence of Laplace transforms. Find the Laplace transform of (13)

(i) $F(t) = (t + 2)e^{-3t} \sin 5t$.

(ii) $G(t) = \begin{cases} t & \text{for } 0 < t < 2 \\ 0 & \text{for } 2 < t < 4 \end{cases}$ and $G(t + 4) = G(t)$.

- (b) Define unit step function. The load distribution on a beam is: (10)

$$p(t) = \begin{cases} 5t, & 0 < t < 4 \\ 20, & t \geq 4 \end{cases}$$

Write $P(t)$ in terms of unit step function and determine the Laplace transform.

- (c) Determine the inverse Laplace transform of (i) $f(s) = \frac{e^{-5s}}{(s-2)^4}$ (12)

(ii) $g(s) = \frac{2s^2 - 4}{(s+1)(s-2)(s-3)}$

6. (a) State the convolution theorem. Use the convolution theorem to evaluate (11)

$$\mathcal{L}^{-1} \left\{ \frac{1}{(s+2)^2(s-2)} \right\}.$$

- (b) The displacement of a building during earthquake satisfies: $x'' + 2x' + x = e^{-t}$ with $x(0) = 0$, $x'(0) = 1$. Find the displacement response using Laplace transform. (13)

- (c) The temperature distribution in a reinforced concrete structure is: (11)

$$T(x, y, z) = x^2y e^{-z} + 3xy^2 + z^3.$$

Find (i) Magnitude of temperature gradient at $(1, 2, 0)$.

(ii) Direction of maximum heat increase.

7. (a) The wind velocity around a high-rise building is: $\vec{V} = (x^2y + yz)\hat{i} + (xy^2 + xz)\hat{j} + (yz^2 + x^2z)\hat{k}$. Determine whether the airflow has expansion or compression at point $(1, 2, 1)$. (11)

- (b) Air velocity field in a smart building system is: $\vec{V} = (x^3 + yz)\hat{i} + (y^3 + xz)\hat{j} + (z^3 + xy)\hat{k}$. Check whether the velocity field is irrotational or not. If exists, find the corresponding velocity potential. (12)

- (c) A force field in a construction system is $\vec{F} = (2xy + z^2)\hat{i} + (x^2 + 2yz)\hat{j} + (2xz + y^2)\hat{k}$. Check whether this field is conservative or not. Find the work done in moving an object in this field from $(1, 0, 0)$ to $(2, 1, 1)$. (12)

8. (a) Evaluate $\iint_s \vec{A} \cdot \vec{n} ds$, where $\vec{A} = 18xz\hat{i} - 12z\hat{j} + 3y\hat{k}$ and s is that part of the plane $2x + 3y + 6z = 12$, which is located in the first octant. (12)

- (b) Evaluate $\iiint_v \vec{F} dv$ where $\vec{F} = 2xz\hat{i} - x\hat{j} + y^2\hat{k}$ and v is the region bounded by the surfaces $x = 0$, $y = 0$, $y = 6$, $z = x^2$, $z = 4$. (11)

- (c) Suppose a force field is given by $\vec{F} = (2x - y + z)\hat{i} + (x + y - z^2)\hat{j} + (3x - 2y + 4z)\hat{k}$. Find the work done in moving a particle once around a circle C in the xy -plane with its centre at the origin and a radius of 3. 12